DSAI4205 – Big Data Analytics (2026/27 Semester One)
Group Project Assignment (25%)

Project Overview:

Your group will receive a randomly assigned business textual dataset. Your objective is to create an effective decision support system that classifies text. The provided dataset is open source, which often means it may contain various hidden issues, such as data inconsistencies and data imbalances. Throughout the project, you should identify any hidden problems and propose solutions for any issues you discover.

The group project can be divided into the following phases.

1. Exploratory Data Analysis (EDA) – Try to understand your data.
   Begin by trying to comprehend what your dataset conveys. Consider the following questions to guide your exploration:
   - What type of data are you currently exploring?
   - What is the granularity of the data?
   - What does the distribution of the data look like? Are there any outliers? Are there any missing or invalid entries?
   - Is there any correlation between the variables you are interested in exploring?
   - How would you cleanly and accurately visualize the relationship among variables?

2. Data Cleaning:
   A crucial step in the data preprocessing pipeline, aimed at improving the quality and accuracy of the dataset before analysis. This process involves several key activities:
   - Identifying Missing Values
   - Handling Outliers
   - Correcting Invalid Entries
   - Removing Duplicates
   - Standardizing Data: Ensure consistency in data representation.
   - Transforming Variables: Apply necessary transformations to variables to enhance their interpretability or suitability for analysis.

   Documenting Changes: Maintain thorough documentation of the cleaning process, including the methods used and any modifications made to the dataset.

3. Data Modeling

   i. Build a logistic regression with TFIDF as the baseline model.
   (Note that TFIDF can only be used in the baseline model)

   ii. Utilize N different data analytics models (such as gradient boosting trees, decision trees, neural networks, etc.) alongside word embedding models (like BERT, GloVe, Word2Vec, etc.). In this phase, it is essential to combine data analytics models with word embedding models.
   Since logistic regression is used in baseline model, logistic regression is not count as one of the N data analytics models. Using a single model to serve as both a data analytics model and a word embedding model is not permitted.
   (Note that in this project, TFIDF is not considered as word embedding model)
   You are required to outline a high-level approach for combining both the embedding model and the analytics model in your project.
   Keep in mind that the scoring will depend on the complexity of the models used, as well as whether you are employing a pre-trained model or training the model from scratch. If a group utilize similar analytics models (for example, decision trees, gradient boosting, random forests), they may receive lower marks due to a lack of diversity in their approach.

   iii. Compare the N data analytics models developed in (ii) with the baseline model. Determine which model performs better and provide justification for the observed differences in performance.

   iv. Improve your baseline models (e.g., ensemble models, fine-tune different hyper-parameter for the models above, etc). After making these improvements, compare the models again with N data analytics models developed in (ii) to determine which one performs better and provide justification for the observed differences in performance.

   v. Visualize with at least N data statistics. There are many ways that can describe patterns found in data, including chart and figures (e.g., bar charts, line graphs, word cloud, word distribution, etc).

   (*N* correspond to your number of group members)

4. Data Visualization (Story telling with data)
   Make use of appropriate visualization to interpret the results to deliver valuable insights
   Data visualization should be applied in EDA, as well as presenting the findings of the project.

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Project Components:

EDA and Data Cleaning (Week 7-8)

Load the dataset and perform initial Exploratory Data Analysis. Define data granularity, check distributions, and identify missing values or outliers. Execute the cleaning pipeline: handle missing values, remove duplicates, and standardize data.

Build the baseline model and prepare Interim Report for the consultation session (Weeks 9-10):

Implement Phase 3i: Build a logistic regression model using TF-IDF. Establish your performance benchmarks (accuracy, F1-score).

Interim Consultation Session: Prepare a summary of your EDA, cleaning steps, and baseline results to discuss. (Schedule a timeslot in week 10/11 Saturday Afternoon, between 2-6pm.)

Advance Modelling + Refinement (Weeks 11-13)

Start **Phase 3ii**: Experiment with N different models (e.g., Random Forest, XGBoost) combined with word embeddings like BERT or Word2Vec. Finalize model selection, perform hyperparameter tuning, and complete the final report.

Final Report Documentation:

- **Technical Deliverable:** All source code, data preprocessing steps, and model implementations must be submitted in a single, well-annotated **Jupyter Notebook (.ipynb)** file.
- **Written Report:** A formal report in **MS Word (.docx)** format is required. This document should synthesize all results, visualizations, and business insights derived throughout the project.
- **Scope:** The report serves as the final, cumulative record of the project, documenting every stage from initial data exploration to advanced modeling and final refinement.

Project Deliverables:

Interim Report and meetup in the consultation session: (20%)

Documents all work performed up to week 9, prepare a summary of your EDA, cleaning steps, and baseline results to discuss.

Submission deadline: 6 Nov 23:55, Week 10

Final Presentation & Final Report (Week 13):

Final Presentation: (20%)

Deliver a 12-minutes presentation showcasing your project.

Assessment criteria:

- Engagement: How well does the team engage the audience? Are they able to capture and maintain interest?

- Clarity and coherence: Is the presentation well-organized and easy to follow?
- Delivery: Are the presenters confident, articulate, and professional?
- Teamwork: How well do team members coordinate with each other during the presentation?
- Response to questions: How effectively does the team respond to questions from the audience?
- A PowerPoint file that you use in the presentation covering highlights in (1) and (3) should be submitted to blackboard by 22 Nov 23:55, Week 13

Final Report: (60%)

Submission deadline: 22 Nov 23:55, Week 13

EDA and Data Cleaning (20%)

- Exploratory Data Analysis (EDA): Successful projects must identify data granularity, distributions, and potential class imbalances.
- Data Cleaning Thoroughness: You are expected to handle missing values, remove duplicates, and standardize data. For text specifically, this includes normalization (lowercasing, punctuation removal) and tokenization (lemmatization or stemming).
- Documentation: A critical component is maintaining a clear log of all modifications made to the dataset.

Baseline Model (10%)

- Implementation: You must build a Logistic Regression model specifically using TF-IDF (Term Frequency-Inverse Document Frequency) for feature extraction.
- Evaluation: The model should be evaluated using standard metrics: Accuracy, Precision, Recall, and F1-score.
- Performance Benchmarking: This model's results must be clearly documented to provide a point of comparison for your more advanced models.

Advance Modelling (30%)

- **Model Selection:** You must use **N different analytics models** (e.g., Gradient Boosting, Neural Networks) paired with **word embedding models** like BERT, GloVe, or Word2Vec.
- Integration Requirement: It is **not feasible to utilize a single model for both tasks** (for instance, employing BERT for both embedding and classification). Instead, you should integrate a separate embedding model with an independent analytics classifier. You are required to outline a high-level approach for combining both the embedding model and the analytics model in your project.

- **Depth of Effort:** Higher scores will be given for more complex models, particularly those not covered in class, as well as for models trained from scratch instead of merely using pre-trained versions.

Refinement (10%)

- Hyperparameter Tuning: Evidence of optimizing parameters to improve accuracy.
- Handling Imbalance: Implementation of strategies to handle the data imbalances identified during EDA.

Uniqueness and Creativity (10%)

- Innovative Feature Engineering: Going beyond standard cleaning to create unique features
- Novel Methodology: Using an unusual combination of models or a unique ensemble technique.
- Insights: Providing exceptionally deep business insights from the classification results that weren't immediately obvious from the raw data.

Project Development & Documentation (20%)

- Thoroughness and clarity of the development documentation.
- Detailed project schedule.
- Clear breakdown of individual’s responsibilities.
- Evidence of experimentation and iterative development/collaboration process.